

3. (a) Geraint is reading a book.



Complete the following sentences by choosing the correct words from the list below. [3]

signals receptors impulses messages neurones

Geraint’s eyes have that detect light reflected from the book.

Information about the light is turned into which travel to the brain
along nerve cells called

(b) Scientists investigated the effect of drinking alcohol on the speed and accuracy of reading a paragraph of text.

The scientists took 24 men and split them into two groups, **A** and **B**.

- The men in group **A** each drank 500 cm³ of beer containing 5 % alcohol.
- The men in group **B** each drank 500 cm³ of non-alcoholic beer.
- Both groups were then given a piece of text to read out loud.
- The time taken to read the text and the number of errors made were recorded.
- The means for each group were then calculated.

The results are shown in **Table 3**.

Table 3

Group	Mean time to complete the reading (s)	Mean number of errors
A	65	3
B	58	0

(i) State **two** conclusions that you can make from the results about the effect of alcohol on reading. [2]

Conclusion 1

.....

Conclusion 2

.....

